

1. Generating the Training Dataset for Skip-Gram Word Embeddings

- The dataset consists of two types of training instances: **positive examples** and **negative examples**
 - o **Positive Examples:**
 - A positive example is a pair $(\mathbf{w}, \mathbf{c}_{pos})$, where:
 - $\mathbf{w}$ is a center (target) word.
 - $\mathbf{c}_{pos}$ is a context word that appears within the context window of $\mathbf{w}$ in the training corpus.
 - Each word in the textual corpus is treated as a center word, and every word appearing within its context window is considered a context word.
 - A positive example is labeled as 1.

Example :

Textual training corpus: "particle swarm optimization is a swarm"

Context window size: 2

The corresponding positive examples are:

(particle, swarm) : 1	(swarm, particle) : 1	(optimization, particle) : 1	(is, swarm) : 1	(a, optimization) : 1	(swarm, is) : 1
(particle, optimization) : 1	(swarm, optimization) : 1	(optimization, swarm) : 1	(is, optimization) : 1	(a, is) : 1	(swarm, a) : 1
	(swarm, is) : 1	(optimization, is) : 1	(is, a) : 1	(a, swarm) : 1	
		(optimization, a) : 1	(is, swarm) : 1		

- o **Negative Examples**
 - For each positive example $(\mathbf{w}, \mathbf{c}_{pos})$, the Skip-gram with Negative Sampling (SGNS) algorithm generates K negative examples.
 - A negative example is a pair $(\mathbf{w}, \mathbf{c}_{neg})$ where:
 - $\mathbf{w}$ is the same center word.
 - $\mathbf{c}_{neg}$ is a noise word, sampled randomly from the vocabulary V .
 - The noise word must be different from both $\mathbf{w}$ and $\mathbf{c}_{pos}$.
 - A negative example is labeled as 0.
 - **Noise word sampling**
 - Noise words are drawn randomly from the vocabulary using:
 - The standard unigram probability:

$$P(\mathbf{c}_{neg}) = \frac{\text{count}(\mathbf{c}_{neg})}{N}$$

Where N is the total number of tokens in the corpus.

- In practice, SGNS uses a weighted unigram distribution:

$$P(\mathbf{c}_{neg}) = \frac{\text{count}(\mathbf{c}_{neg})^\alpha}{\sum_{\mathbf{w}} (\text{count}(\mathbf{w}))^\alpha}$$

With $\alpha = 0.75$, which increases the probability of sampling rare words.

Année Universitaire : 2025/2026 Master 2 : SII Module : TALN	Université des Sciences et de la Technologie Houari Boumediene Faculté d'Informatique Département d'Intelligence Artificielle et Sciences des Données	TP N°8 Word Embedding Part 1
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- To sample a noise word:
 - (i) Compute the unigram probabilities for all vocabulary words.
 - (ii) Build a cumulative distribution vector of size $|V|$ (its last element equals 1).
 - (iii) Draw a random number between 0 and 1.
 - (iv) Select the vocabulary entry whose cumulative probability interval contains this value.

Example

Textual training corpus: "**particle swarm optimization is a swarm**"

Context window size: 2

$K = 2$

The corresponding positive examples are:

For the positive example: (particle, swarm) : 1

We generate the following negative examples: (particle, is) : 0 (particle, a) : 0

2. Implementation :

Displaying all context words for a given center (target) word

☒ Content of one volume
 ☐ Content of all articles

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☐ Normalization

Porter Stemmer ▼

Center	Context
pso	optimization
pso	</s>
pso	swarm
pso	<s>
pso	standard
pso	algorithms
pso	and
pso	</s>
pso	optimization
pso	face

Parameters

Window size

2

Negative examples (k)

5

Embedding dimension

30

Alpha(α)

0.75

Display

(Centre, Context)

Word center

pso

One-Center

All-Centers

All-Dataset

Displaying all context words for each center (target) word

Content of one volume

Content of all articles

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Normalization

Porter Stemmer

Center	Context
<s>	explosive
the	<s>
the	explosive
the	</s>
the	growth
explosive	the
explosive	growth
explosive	<s>
explosive	of
growth	explosive

Parameters

Window size

2

Negative examples (k)

5

Embedding dimension

30

Alpha(α)

0.75

Display

(Centre, Context)

Word center

pso

One-Center

All-Centers

All-Dataset

Displaying the whole training dataset

☒ Content of one volume
 ☐ Content of all articles

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Normalization

Porter Stemmer

Center	Context/noise word	Label
advanced	balancing	0
advanced	temporal	0
data	advanced	1
data	algorithm	0
data	like	0
data	systems	0
data	windows	0
data	paid	0
data	mining	1
data	research	0
data	metrics	0
data	</s>	0
data	diseases	0
data	and	0
data	of	1
data	shoulder	0

Parameters

Window size

Negative examples (k)

2

5

Embedding dimension

Alpha(α)

30

0.75

Display

(Centre, Context)

Word center

pso

One-Center

All-Centers

All-Dataset